THE SPECTRUM OF THE TOP-TO-RANDOM CHAIN ON TWO-COLOURED WORDS

ABSTRACT. Let Ω_n be the set of words containing n letters $\mathtt{R}$ and n letters $\mathtt{G}$. At each step the leftmost letter is removed and reinserted, uniformly at random, into one of the $2n$ slots determined by the remaining $2n-1$ letters. We show that the transition matrix is diagonalizable, that its eigenvalues are 1 and $j/(2n)$ for $0 \le j \le 2n-2$, and that $j/(2n)$ has multiplicity equal to the number of words of length $2n-2-j$ in $\mathtt{R},\mathtt{G}$ with at most $n-1$ copies of each letter. The proof lumps the chain from the top-to-random shuffle of $2n$ distinct cards and determines the $\mathfrak{S}_{2n}$-module structure of the eigenspaces of that shuffle by an equivariant filtration in the spirit of Brown's theory of left-regular bands. We work out the cases $n \le 6$ in detail; give equivalent forms of the multiplicities, a Pascal-type recurrence and a generating function; prove that the multiplicities in the lower half of the spectrum have no hypergeometric closed form; split the spectrum along the Specht modules $S^{(2n-i,i)}$; prove that the colour-symmetric and colour-antisymmetric parts have the same spectrum apart from the two largest eigenvalues; and examine a proposed binomial formula for the multiplicities.

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1. The chain and the main result

1.1. **The chain.** Fix $n \geq 1$ and put $N = 2n$. Let Ω_n be the set of words of length N in the letters $\mathtt{R}$ and $\mathtt{G}$ that contain n of each, so $|\Omega_n| = \binom{2n}{n}$. One step of the chain removes the leftmost letter of the current word. The remaining $N-1$ letters determine N slots (before the first letter, between consecutive letters, and after the last), and the removed letter is inserted into one of them, chosen uniformly at random. Thus

$$P(w, w') = \frac{1}{N} \#\{\text{slots whose use turns } w \text{ into } w'\}.$$

For example, from $\mathtt{RGRG}$ we remove $\mathtt{R}$ and are left with $\mathtt{GRG}$; the four insertions give $\mathtt{RGRG}$, $\mathtt{GRRG}$, $\mathtt{GRRG}$, $\mathtt{GRGR}$, so $P(\mathtt{RGRG}, \cdot\,)$ puts mass $\frac{1}{4}, \frac{1}{2}, \frac{1}{4}$ on $\mathtt{RGRG}$, $\mathtt{GRRG}$, $\mathtt{GRGR}$.

The matrix P acts on functions $f\colon \Omega_n \to \mathbb{C}$ by $(Pf)(w) = \sum_{w'} P(w, w') f(w')$. Inserting the removed letter into the p-th slot moves the first letter to position p. Regarding a word as a map $w\colon [N] \to \{\mathtt{R}, \mathtt{G}\}$ from positions to letters, this says

$$(Pf)(w) = \frac{1}{N} \sum_{p=1}^{N} f(w \circ \pi_p), \qquad \pi_p(q) = \begin{cases} q+1, & q < p, \\ 1, & q = p, \\ q, & q > p, \end{cases} \tag{1.1}$$

with $\pi_p \in \mathfrak{S}_N$ a permutation of positions. We use this description in Sections 5 and 7.

1.2. **Conventions.** Throughout, j is the numerator of an eigenvalue j/N and

$$L = N - 2 - j.$$

Binomial coefficients $\binom{L}{i}$ with $L \geq 0$ vanish unless $0 \leq i \leq L$. We write $B(L, k) = \sum_{i=0}^{k} \binom{L}{i}$ for partial row sums of Pascal's triangle, d_m for the number of derangements of m objects, and $C_n = \binom{2n}{n}/(n+1)$ for the Catalan numbers.

1.3. **The main result.**

Theorem 1.1. *The matrix P is diagonalizable. Its eigenvalues are 1, with multiplicity 1, and j/N for $0 \leq j \leq N-2$, with multiplicity*

$$m_n(j) = \sum_{i=\max(0,\, L-n+1)}^{\min(L,\, n-1)} \binom{L}{i}, \qquad L = N - 2 - j, \tag{1.2}$$

which is the number of words of length L in $\mathtt{R}$ and $\mathtt{G}$ that contain at most $n-1$ copies of each letter. In particular $(N-1)/N$ is not an eigenvalue. Setting $m_n(N-1) = 0$ and $m_n(N) = 1$, we have for every $0 \leq j \leq N$

$$m_n(j) = \sum_{\substack{r+g=j \\ 0 \leq r,\, g \leq n}} D(n-r,\, n-g), \qquad D(a, b) = \begin{cases} \binom{a+b-2}{a-1}, & a, b \geq 1, \\ 1, & a = b = 0, \\ 0, & \textit{otherwise.} \end{cases} \tag{1.3}$$

The multiplicities for $n \le 6$ are listed in Table 1.1; they agree with the spectra computed directly from the matrices (Sections 3 and 9).

TABLE 1.1. Multiplicities of the eigenvalues $N/N, (N-1)/N, \dots, 1/N, 0$, from the top.

n	$\lvert\Omega_n\rvert$	multiplicities of $N/N,\ (N-1)/N,\ \dots,\ 0/N$
1	2	1, 0, 1
2	6	1, 0, 1, 2, 2
3	20	1, 0, 1, 2, 4, 6, 6
4	70	1, 0, 1, 2, 4, 8, 14, 20, 20
5	252	1, 0, 1, 2, 4, 8, 16, 30, 50, 70, 70
6	924	1, 0, 1, 2, 4, 8, 16, 32, 62, 112, 182, 252, 252

Remark 1.2 (Distinct cards). For the top-to-random shuffle of N distinct cards the eigenvalues are again j/N, $j \in \{0, \dots, N-2, N\}$, and j/N has multiplicity $\binom{N}{j} d_{N-j}$, the number of permutations of $[N]$ with exactly j fixed points [BHR99, BD98, Br00, DFP92]. The value $(N-1)/N$ is missing for the same reason as here: $d_1 = 0$. Formula (1.3) is the coloured analogue of $\sum_{|X|=j} d_{N-j}$: subsets X of the cards are replaced by sub-multisets with r red and g green cards, and the derangement number d_{N-j} of the remaining cards is replaced by the *coloured derangement number* $D(a,b)$ of the remaining a red and b green cards. For $(a,b) \neq (0,0)$, $D(a,b)$ is the number of words with a letters `R` and b letters `G` that begin with `R` and end with `G`.

Remark 1.3 (Plan). Section 2 makes the colour-swap symmetry precise, and Section 3 works out $n = 1, 2, 3, 5$. Section 4 gives equivalent forms of (1.2). Section 5 proves Theorem 1.1 in three steps: the chain is a lumping of the distinct-card shuffle (Proposition 5.1); the $\mathfrak{S}_N$-module structure of the eigenspaces of the distinct-card shuffle is determined by an equivariant filtration (Theorem 5.5); and Frobenius reciprocity counts the invariants. The same argument handles decks of arbitrary content (Corollary 5.9). Section 6 discusses how far (1.2) simplifies, including a proof that the lower half of the spectrum has no hypergeometric closed form (Theorem 6.5). Section 7 splits the spectrum along Specht modules and proves the colour-swap theorem (Theorem 7.4). Section 8 examines a proposed binomial formula, and Section 9 lists the computer checks; the code is in Section A.

2. THE COLOUR-SWAP SYMMETRY

Let $\sigma\colon \Omega_n \to \Omega_n$ exchange `R` and `G` in every position, so for instance $\sigma(\texttt{RRGRGG}) = \texttt{GGRGRR}$.

- σ is an involution without fixed points (it changes the first letter), so Ω_n splits into $\binom{2n}{n}/2$ pairs $\{x, \sigma x\}$.
- The chain does not see the names of the colours: $P(\sigma x, \sigma y) = P(x, y)$.

Let $(Sf)(x) = f(\sigma x)$. Then $S^2 = I$ and $SP = PS$. Put

$$V_+ = \{f :\ f(\sigma x) = f(x) \text{ for all } x\}, \qquad V_- = \{f :\ f(\sigma x) = -f(x) \text{ for all } x\}.$$

Proposition 2.1. *The two parts have the following properties.*

(i) $\mathbb{C}^{\Omega_n} = V_+ \oplus V_-$, *via* $f = \frac{1}{2}(f + Sf) + \frac{1}{2}(f - Sf)$.
(ii) V_+ *and* V_- *are* P*-invariant.*
(iii) *Choose a representative* x *in each pair* $O = \{x, \sigma x\}$. *The functions* $\mathbf{1}_x + \mathbf{1}_{\sigma x}$ *form a basis of* V_+ *and the functions* $\mathbf{1}_x - \mathbf{1}_{\sigma x}$ *a basis of* V_-*; in particular* $\dim V_\pm = \binom{2n}{n}/2$.
(iv) *In these bases* P *acts on* $V_\pm$ *by the matrices*

$$P_\pm(O, O') = P(x, y) \pm P(x, \sigma y),$$

where x and y are the chosen representatives of O and O'. Hence the spectrum of P is the union, with multiplicities, of the spectra of P_+ and P_-.

Proof. (i) and (ii) follow from $S^2 = I$ and $SP = PS$, and (iii) is immediate. For (iv), if $f = \sum_{O'} a_{O'}(\mathbf{1}_y \pm \mathbf{1}_{\sigma y})$ then $(Pf)(x) = \sum_{O'} a_{O'}\big(P(x,y) \pm P(x,\sigma y)\big)$, and $(Pf)(x)$ is the coefficient of $\mathbf{1}_x \pm \mathbf{1}_{\sigma x}$ in Pf. □

The matrix P_+ is the transition matrix of the *lumped* chain on the pairs $\{x, \sigma x\}$, that is, of the chain observed only up to colour swap (for instance, by recording at each position whether its letter agrees with the first letter); it is a genuine Markov chain because P commutes with σ. The matrix P_- is the signed version and is not stochastic. The constant function lies in V_+, so the eigenvalue 1 belongs to P_+. Theorem 7.4 below shows that P_+ and P_- have the same spectrum, except that 1 belongs to P_+ only and $(N-2)/N$ to P_- only.

3. Small cases

3.1. $n = 1$. Here $\Omega_1 = \{\mathtt{RG}, \mathtt{GR}\}$. From either state, removing the first letter leaves a single letter, and the two insertions give RG and GR. So $P = \frac{1}{2}\left(\begin{smallmatrix} 1 & 1 \\ 1 & 1 \end{smallmatrix}\right)$, with eigenvalue 1 (constant eigenvector, in V_+) and 0 (eigenvector $(1,-1)$, in V_-). The value $(N-1)/N = \frac{1}{2}$ does not occur.

3.2. $n = 2$. Number the six states

$$1 = \mathtt{RRGG}, \quad 2 = \mathtt{RGRG}, \quad 3 = \mathtt{RGGR}, \quad 4 = \mathtt{GRRG}, \quad 5 = \mathtt{GRGR}, \quad 6 = \mathtt{GGRR}.$$

Removing the first letter and reinserting it into each of the four slots in turn gives

$$4P = \begin{pmatrix} 2 & 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 2 & 1 & 0 \\ 0 & 0 & 1 & 0 & 1 & 2 \\ 2 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 2 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 1 & 2 \end{pmatrix}.$$

For instance, from RGRG we are left with GRG, and the insertions give RGRG, GRRG, GRRG, GRGR, which is the second row.

The colour swap pairs $1 \leftrightarrow 6$, $2 \leftrightarrow 5$, $3 \leftrightarrow 4$.

Symmetric part. Functions in V_+ have the form $f = (a,b,c,c,b,a)$, and $4P$ sends (a,b,c) to $(2a+b+c,\ 2b+2c,\ 2a+b+c)$. So

$$4P_+ = \begin{pmatrix} 2 & 1 & 1 \\ 0 & 2 & 2 \\ 2 & 1 & 1 \end{pmatrix}.$$

Its trace is 5; its rows sum to 4, so 4 is an eigenvalue; its first and third rows are equal, so 0 is an eigenvalue; and the trace forces the third eigenvalue to be 1. Thus P_+ has eigenvalues $1, \frac{1}{4}, 0$.

Antisymmetric part. Functions in V_- have the form $f = (a,b,c,-c,-b,-a)$, and $4P$ sends (a,b,c) to $(2a+b+c,\ -2c,\ -2a-b+c)$. So

$$4P_- = \begin{pmatrix} 2 & 1 & 1 \\ 0 & 0 & -2 \\ -2 & -1 & 1 \end{pmatrix},$$

with characteristic polynomial $x(x-1)(x-2)$. Thus P_- has eigenvalues $\frac{1}{2}, \frac{1}{4}, 0$.

The spectrum of P is therefore $1, \frac{1}{2}, \frac{1}{4}, \frac{1}{4}, 0, 0$, as Theorem 1.1 predicts: for $j = 0,1,2$ we have $L = 2,1,0$, and the words of length L with at most one copy of each letter are RG and GR; R and G;

and the empty word. The value $\frac{3}{4}$ does not occur. Right eigenvectors (functions on the states, listed in the order above) are given in Table 3.1.

TABLE 3.1. Eigenvectors for $n = 2$.

eigenvalue	eigenvector f	part
1	$(1, 1, 1, 1, 1, 1)$	V_+
1/2	$(0, 1, -1, 1, -1, 0)$	V_-
1/4	$(1, -2, 1, 1, -2, 1)$	V_+
1/4	$(1, -2, 1, -1, 2, -1)$	V_-
0	$(0, 1, -1, -1, 1, 0)$	V_+
0	$(1, -2, 0, 0, 2, -1)$	V_-

3.3. $n = 3$. There are $\binom{6}{3} = 20$ states. Each row of $6P$ is obtained by removing the first letter and reinserting it into each of the 6 slots; three sample rows are

from	to (entry of $6P$)
`RRRGGG`	`RRRGGG` (3), `RRGGGR` (1), `RRGGRG` (1), `RRGRGG` (1)
`RGRGRG`	`GRGRGR` (1), `GRGRRG` (2), `GRRGRG` (2), `RGRGRG` (1)
`RRGGGR`	`RGGGRR` (2), `RGGRGR` (1), `RGRGGR` (1), `RRGGGR` (2)

The eigenvalues of $6P$ are integers. Computing $\dim\ker(6P - jI)$ exactly, by integer row reduction, for each j gives the multiplicities below; they add up to 20, so P is diagonalizable with spectrum

$$1, \quad \tfrac{2}{3}, \quad \tfrac{1}{2}\ (\times 2), \quad \tfrac{1}{3}\ (\times 4), \quad \tfrac{1}{6}\ (\times 6), \quad 0\ (\times 6),$$

accounting for all 20 states. Table 3.2 compares this with Theorem 1.1.

TABLE 3.2. $n = 3$: the eigenvalue $j/6$ and the words of length $L = 4 - j$ with at most two copies of each letter.

eigenvalue	j	L	words counted	multiplicity
1	6	—	—	1
5/6	5	—	(never occurs)	0
2/3	4	0	the empty word	1
1/2	3	1	`R`, `G`	2
1/3	2	2	all 4 words	4
1/6	1	3	8 words minus `RRR`, `GGG`	6
0	0	4	16 words minus the 10 with ≥ 3 copies of a letter	6

The colour swap splits the 20-dimensional space into two 10-dimensional parts, and the eigenvalues of $6P_\pm$ are

$$6P_+ : \ 6, 3, 2, 2, 1, 1, 1, 0, 0, 0, \qquad 6P_- : \ 4, 3, 2, 2, 1, 1, 1, 0, 0, 0.$$

The two lists agree except at the top: the symmetric part contains the stationary eigenvalue 1 and the antisymmetric part contains $\frac{2}{3}$, the second largest eigenvalue. The same happens for $n = 2$, where the parts are $\{1, \frac{1}{4}, 0\}$ and $\{\frac{1}{2}, \frac{1}{4}, 0\}$; Theorem 7.4 proves this pattern for all n.

TABLE 3.3. $n = 5$: the formula (1.2) with $L = 8 - j$, and the computed multiplicities.

eigenvalue	j	L	$\sum_i \binom{L}{i}$	formula	computed
1	10	—	—	1	1
9/10	9	—	—	0	0
8/10	8	0	$\binom{0}{0}$	1	1
7/10	7	1	2^1	2	2
6/10	6	2	2^2	4	4
5/10	5	3	2^3	8	8
4/10	4	4	2^4	16	16
3/10	3	5	$5 + 10 + 10 + 5$	30	30
2/10	2	6	$15 + 20 + 15$	50	50
1/10	1	7	$\binom{7}{3} + \binom{7}{4}$	70	70
0	0	8	$\binom{8}{4}$	70	70

3.4. $n = 5$. There are 252 states. Table 3.3 lists the multiplicities, both as predicted by (1.2) and as found from the matrix.

The computed column was obtained in four independent ways.

(i) *Floating-point eigenvalues.* The 252 eigenvalues of P were computed with a standard dense eigensolver. Every value of 10λ lies within 3.2×10^{-13} of an integer, every imaginary part is below 2.2×10^{-14}, and counting the eigenvalues at each integer gives the formula column.

(ii) *Exact eigenspace dimensions.* Since $10P$ has integer entries, the dimensions of the kernels of $10P - jI$ and of $(10P - jI)^2$ can be computed exactly, by integer row reduction. For every j the two numbers coincide and equal $m_5(j)$, and they add up to 252. Hence $10P$ is diagonalizable, its only eigenvalues are $0, 1, \ldots, 8$ and 10, and the multiplicities are those of Table 3.3. This proves Theorem 1.1 for $n = 5$ independently of Section 5.

(iii) *Singular values.* For each j, the singular values of $10P - jI$ separate cleanly: those counted as zero are at most 4.1×10^{-15}, and the next smallest is at least 0.139. The resulting kernel dimensions agree with (ii).

(iv) *Traces of powers.* In exact integer arithmetic, $\operatorname{tr}\big((10P)^k\big) = \sum_j m_5(j)\, j^k$ for every $0 \le k \le 18$. For example the common values are 420, 1402, 247830, 11977281982 and 10217090942574512 62 for $k = 1, 2, 5, 10, 18$.

4. Equivalent forms of the multiplicities

Proposition 4.1. *Let $0 \le j \le N - 2$ and $L = N - 2 - j$. Each of the following equals $m_n(j)$.*

(a) *The number of words of length L in* `R`, `G` *with at most $n-1$ letters* `R` *and at most $n-1$ letters* `G`.

(b) *The number of words of length L in which the numbers of* `R`*'s and* `G`*'s differ by at most j:*

$$m_n(j) = \sum_{|2i - L| \le j} \binom{L}{i} = 2^L \Pr\big(|S_L| \le j\big),$$

where S_L is the position of a simple ± 1 random walk after L steps.

(c) $2B(L, n-1) - 2^L = 2\sum_{i=0}^{n-1} \binom{L}{i} - 2^L$.

(d) $S(L)$, *where* $S(0) = 1$ *and* $S(\ell) = 2S(\ell - 1) - 2\binom{\ell - 1}{n-1}$ *for* $1 \le \ell \le L$.

(e) $\sum_{r+g=j} D(n - r, n - g)$, *with* $0 \le r, g \le n$ *and D as in (1.3). This also holds for $j = N - 1$ and $j = N$, with values 0 and 1.*

Proof. (a) A word with i letters $\mathtt{R}$ has $L-i$ letters $\mathtt{G}$, and the conditions $i \le n-1$ and $L-i \le n-1$ say $L-n+1 \le i \le n-1$, which is the range of summation in (1.2).

(b) The two conditions say $2i - L \le 2n-2-L = j$ and $L-2i \le j$.

(c) By the symmetry $\binom{L}{i} = \binom{L}{L-i}$, $\sum_{i \le L-n}\binom{L}{i} = \sum_{i \ge n}\binom{L}{i} = 2^L - B(L,n-1)$, so the sum over $L-n+1 \le i \le n-1$ equals $B(L,n-1) - \big(2^L - B(L,n-1)\big)$.

(d) Let $S(\ell)$ be the number of words of length ℓ with at most $n-1$ copies of each letter. Extending each such word of length $\ell-1$ by one letter gives $2S(\ell-1)$ words of length ℓ. An extension is invalid exactly when it adds an n-th copy of a letter, and for each letter there are $\binom{\ell-1}{n-1}$ such extensions (the other letter then occurs $\ell - n \le n-1$ times, since $\ell \le L \le 2n-2$).

(e) Let $j \le N-2$. A term with $r = n$ or $g = n$ vanishes, because $D(0,b) = D(a,0) = 0$ for $a, b \ge 1$ and $r = g = n$ would force $j = N$. For $r, g \le n-1$ we have $D(n-r, n-g) = \binom{2n-2-j}{n-r-1} = \binom{L}{n-1-r}$. As r runs over $\max(0, j-n+1) \le r \le \min(j, n-1)$, the index $i = n-1-r$ runs over $\max(0, n-1-j) \le i \le \min(n-1, L)$, and $n-1-j = L-n+1$, so this is the range in (1.2). For $j = N$ only $(r,g) = (n,n)$ occurs, giving $D(0,0) = 1$; for $j = N-1$ the terms are $D(0,1) = D(1,0) = 0$. □

When $L \le n-1$, that is $j \ge n-1$, no word of length L contains n copies of a letter, so $m_n(j) = 2^L$; this and other simplifications are discussed in Section 6.

5. Proof of the main theorem

5.1. **Symmetric functions.** We use the ring Λ of symmetric functions with the complete, elementary, power sum and Schur functions h_k, e_k, $p_1 = h_1 = e_1$ and s_λ, and the convention $h_k = e_k = 0$ for $k < 0$. Put $H(t) = \sum_{k \ge 0} h_k t^k$ and $E(t) = \sum_{k \ge 0} e_k t^k$, so that $H(t)E(-t) = 1$. The Frobenius characteristic ch sends a representation of $\mathfrak{S}_N$ to Λ^N; it satisfies $\operatorname{ch}\operatorname{Ind}_{\mathfrak{S}_\mu}^{\mathfrak{S}_N} \mathbf{1} = h_\mu$, $\operatorname{ch}(\operatorname{sgn}_N) = e_N$, $\operatorname{ch} S^\lambda = s_\lambda$ and $\operatorname{ch}\operatorname{Ind}_{\mathfrak{S}_a \times \mathfrak{S}_b}^{\mathfrak{S}_{a+b}}(V \otimes W) = \operatorname{ch} V \cdot \operatorname{ch} W$, and $\langle \operatorname{ch} V, \operatorname{ch} W\rangle = \dim \operatorname{Hom}_{\mathfrak{S}_N}(V, W)$ for the Hall inner product [Mac95, Sag01].

For an operator A on a finite-dimensional space V we write $V_\lambda(A) = \ker(A-\lambda)^{\dim V}$ for the generalized eigenspace. If a group acts on V commuting with A, each $V_\lambda(A)$ is a submodule, because the projection onto $V_\lambda(A)$ along the other generalized eigenspaces is a polynomial in A. For the same reason, a short exact sequence $0 \to V' \to V \to V'' \to 0$ of modules with compatible operators gives short exact sequences $0 \to V'_\lambda \to V_\lambda \to V''_\lambda \to 0$, so characters of generalized eigenspaces add along filtrations.

5.2. **Step 1: lumping.** An *arrangement* of N distinct cards is a bijection $c\colon [N] \to [N]$ from positions to cards; $c(1)$ is the top card. The group $\mathfrak{S}_N$ acts on arrangements by relabelling the cards, $c \mapsto g \circ c$, and on functions by $(g \cdot f)(c) = f(g^{-1} \circ c)$; as a module, $\mathbb{C}^{\mathfrak{S}_N}$ is the regular representation, with characteristic h_1^N. The *top-to-random* operator is

$$(Tf)(c) = \frac{1}{N}\sum_{p=1}^{N} f(c \circ \pi_p),$$

where $c \circ \pi_p$ is the arrangement obtained from c by moving the top card to position p, with π_p as in (1.1). Relabelling acts on the left and moves act on the right, so T commutes with relabelling. The *random-to-top* operator is $(Rf)(c) = \frac{1}{N}\sum_{i=1}^{N} f(i \cdot c)$, where $i \cdot c$ is obtained from c by moving card i to the top. Moving the top card to position p is undone by moving the same card back to the top, so $R(c,c') = T(c',c)$, that is $R = T^{\mathsf{T}}$; R also commutes with relabelling.

Colour cards $1, \dots, n$ red and $n+1, \dots, N$ green, by $\kappa\colon [N] \to \{\mathtt{R}, \mathtt{G}\}$, and put $\varphi(c) = \kappa \circ c \in \Omega_n$, the colour word of c. Let $\Gamma = \{g \in \mathfrak{S}_N : \kappa \circ g = \kappa\} = \mathfrak{S}_n \times \mathfrak{S}_n$. The fibres of φ are exactly the Γ-orbits.

Proposition 5.1. *For every $f\colon \Omega_n \to \mathbb{C}$ we have $T(f\circ\varphi) = (Pf)\circ\varphi$. The map $f \mapsto f\circ\varphi$ identifies $\mathbb{C}^{\Omega_n}$ with the space $(\mathbb{C}^{\mathfrak{S}_N})^{\Gamma}$ of Γ-invariant functions and intertwines P with T. Consequently, for every $\lambda \in \mathbb{C}$ the algebraic multiplicity of λ as an eigenvalue of P equals $\dim\big(V_\lambda(T)\big)^{\Gamma}$, where $V = \mathbb{C}^{\mathfrak{S}_N}$.*

Proof. $(f\circ\varphi)(c\circ\pi_p) = f(\kappa\circ c\circ\pi_p) = f(\varphi(c)\circ\pi_p)$; averaging over p and using (1.1) gives $(Pf)(\varphi(c))$. A function on arrangements is Γ-invariant exactly when it is constant on the fibres of φ, that is, of the form $f\circ\varphi$. Finally, the projection onto $V_\lambda(T)$ is a polynomial in T, hence commutes with averaging over Γ, so the generalized λ-eigenspace of T restricted to V^{Γ} is $V_\lambda(T)\cap V^{\Gamma} = V_\lambda(T)^{\Gamma}$. □

5.3. **Injective words and the prefix filtration.**

Definition 5.2. An *injective word* of length k over $[N]$ is a sequence $u = u_1u_2\cdots u_k$ of *distinct* elements of $[N]$; equivalently, an injective map $[k]\to[N]$, $i\mapsto u_i$. Over the alphabet [4], for example, 3 1 4 is an injective word of length 3 and 3 1 3 is not. There are $N!/(N-k)!$ injective words of length k; the empty word is the injective word of length 0, and the injective words of length N are the arrangements. (In other terminology, injective words of length k are k-permutations or partial permutations.)

Remark 5.3. With the product "concatenate, then delete every letter that has already appeared", for instance $(3\,1)\cdot(1\,4\,2) = 3\,1\,4\,2$, injective words form the free left-regular band. Its random walk on the chambers (the arrangements) driven by the one-letter words is random-to-top; this is how the Tsetlin library enters Brown's theory [Br00].

Let F_k be the space of functions on injective words of length k, with $\mathfrak{S}_N$ acting by relabelling. The stabilizer of a word is the group of permutations fixing each of its k letters, so

$$F_k \cong \operatorname{Ind}_{\mathfrak{S}_1^k\times\mathfrak{S}_{N-k}}^{\mathfrak{S}_N}\mathbf{1}, \qquad \operatorname{ch} F_k = h_1^k\,h_{N-k}.$$

Let $U_k \subseteq \mathbb{C}^{\mathfrak{S}_N}$ be the space of functions f such that $f(c)$ depends only on the top k cards $(c(1),\dots,c(k))$. The top k cards form an injective word of length k, and every such word occurs, so $U_k \cong F_k$ via $f(c) = \tilde f(c(1)\cdots c(k))$. This gives a filtration by submodules

$$U_0 \subset U_1 \subset \cdots \subset U_{N-1} = U_N = \mathbb{C}^{\mathfrak{S}_N},$$

where U_0 is the space of constants and $U_{N-1} = U_N$ because the top $N-1$ cards determine the arrangement.

Lemma 5.4. *Each U_k is invariant under R. For $1\le k\le N$ and $f\in U_k$,*

$$NRf \equiv A_k f \pmod{U_{k-1}}, \qquad (A_k\tilde f)(u) = \sum_{t=1}^{k-1}\tilde f\big(u_t\,u_1\cdots\widehat{u_t}\cdots u_k\big),$$

that is, A_k moves one of the first $k-1$ letters to the front and leaves the last letter u_k in place (the hat denotes omission; $A_1 = 0$). The operator A_k preserves the functions on k-words that depend only on the first $k-1$ letters, and acts on them as unnormalized random-to-top on $(k-1)$-words.

Proof. Let $u = c(1)\cdots c(k)$. If $i = u_t$ with $t\le k$, the top k cards of $i\cdot c$ are $u_t\,u_1\cdots\widehat{u_t}\cdots u_k$; if $i\notin\{u_1,\dots,u_k\}$, they are $i\,u_1\cdots u_{k-1}$. Hence

$$N(Rf)(c) = \sum_{t=1}^{k}\tilde f\big(u_t\,u_1\cdots\widehat{u_t}\cdots u_k\big) + \sum_{i\notin\{u_1,\dots,u_k\}}\tilde f(i\,u_1\cdots u_{k-1}), \tag{5.1}$$

which depends only on u; so $Rf\in U_k$. Write the second sum as

$$\sum_{i\notin\{u_1,\dots,u_{k-1}\}}\tilde f(i\,u_1\cdots u_{k-1}) \;-\; \tilde f(u_k\,u_1\cdots u_{k-1}).$$

The first term depends only on $u_1 \cdots u_{k-1}$ and so lies in U_{k-1}; the subtracted term cancels the term $t = k$ of the first sum in (5.1). This proves the congruence. If $\tilde{f}(u)$ depends only on $u_1 \cdots u_{k-1}$, then moving u_t ($t \le k-1$) to the front rearranges the first $k-1$ letters, so $A_k \tilde{f}$ again depends only on them, through unnormalized random-to-top on $(k-1)$-words. □

5.4. **Step 2: the eigenspaces of the distinct-card shuffle.**

Theorem 5.5. *For $N \ge 0$ and $\ell \in \mathbb{Z}$ let $c_N(\ell) \in \Lambda^N$ be the Frobenius characteristic of the generalized eigenspace of NR on $\mathbb{C}^{\mathfrak{S}_N}$ for the eigenvalue ℓ, with $\mathfrak{S}_N$ acting by relabelling. Then*

$$c_N(\ell) = h_\ell \, \delta_{N-\ell} \quad (0 \le \ell \le N), \qquad c_N(\ell) = 0 \text{ otherwise},$$

where

$$\sum_{m \ge 0} \delta_m t^m = \frac{E(-t)}{1 - h_1 t}, \qquad \text{that is,} \qquad \delta_m = \sum_{i=0}^{m} (-1)^i e_i \, h_1^{m-i}.$$

The same holds for NT. In particular NR and NT have only the eigenvalues $0, 1, \dots, N-2$ and N, and the eigenvalue ℓ has multiplicity $\binom{N}{\ell} d_{N-\ell}$.

Proof. The pieces. For $1 \le k \le N$ let $Q_k = U_k / U_{k-1}$. By Lemma 5.4, NR induces on Q_k the same operator that A_k induces on F_k / F_{k-1}, where F_{k-1} is embedded in F_k as the functions of the first $k-1$ letters. So the ℓ-part of Q_k has characteristic equal to that of (F_k, A_k) minus that of $(F_{k-1}, A_k|_{F_{k-1}})$.

The space F_k is the direct sum, over k-subsets $X \subseteq [N]$ and $s \in X$, of the functions on injective words with letter set X and last letter s. The operator A_k preserves each summand and acts on it as unnormalized random-to-top on the orderings of $X \setminus \{s\}$. Relabelling permutes the summands transitively, with stabilizer $\mathfrak{S}_{X\setminus\{s\}} \times \mathfrak{S}_{\{s\}} \times \mathfrak{S}_{[N]\setminus X} \cong \mathfrak{S}_{k-1} \times \mathfrak{S}_1 \times \mathfrak{S}_{N-k}$. Hence (F_k, A_k) is induced from $(\mathbb{C}^{\mathfrak{S}_{k-1}}, (k-1)R) \otimes \mathbf{1} \otimes \mathbf{1}$, and its ℓ-part has characteristic $c_{k-1}(\ell)\, h_1 h_{N-k}$. In the same way, $(F_{k-1}, A_k|_{F_{k-1}})$ is induced from $\mathfrak{S}_{k-1} \times \mathfrak{S}_{N-k+1}$, and its ℓ-part has characteristic $c_{k-1}(\ell)\, h_{N-k+1}$.

The recursion. On U_0, the constants, NR acts as N. Adding up the pieces of the filtration gives, for all $N \ge 0$,

$$c_N(\ell) = [\ell = N]\, h_N + \sum_{k=1}^{N} \big(h_1 h_{N-k} - h_{N-k+1}\big)\, c_{k-1}(\ell), \tag{5.2}$$

so in particular $c_0(\ell) = [\ell = 0]$. (For $k = N$ the coefficient vanishes, matching $U_N = U_{N-1}$.)

Solving. Let $C_\ell(t) = \sum_{N \ge 0} c_N(\ell) t^N$. Multiplying (5.2) by t^N and summing,

$$C_\ell = h_\ell t^\ell + t\Big(h_1 H(t) - \frac{H(t) - 1}{t}\Big) C_\ell = h_\ell t^\ell + \big(1 - H(t) + t\, h_1 H(t)\big) C_\ell,$$

so $C_\ell \, H(t)(1 - h_1 t) = h_\ell t^\ell$ and $C_\ell = h_\ell t^\ell E(-t)/(1 - h_1 t) = h_\ell t^\ell \sum_m \delta_m t^m$. Reading off the coefficient of t^N gives the formula for $c_N(\ell)$.

Dimensions. A representation with characteristic $F \in \Lambda^m$ has dimension $\langle F, h_1^m \rangle$; since $\dim \mathrm{Ind}_{\mathfrak{S}_i \times \mathfrak{S}_1^{m-i}} (\mathrm{sgn} \otimes \mathbf{1}) = m!/i!$, we get $\dim \delta_m = \sum_i (-1)^i m!/i! = d_m$ and $\dim h_\ell \delta_{N-\ell} = \binom{N}{\ell} d_{N-\ell}$. Also $\delta_1 = h_1 - e_1 = 0$.

The operator T. $T = R^{\mathsf{T}}$ is the adjoint of R for the standard inner product, which is invariant under relabelling. Hence $V_\ell(NT)$ is the orthogonal complement of $\bigoplus_{\ell' \ne \ell} V_{\ell'}(NR)$, which is isomorphic as a module to the dual of $V_\ell(NR)$. Characters of $\mathfrak{S}_N$ are real, so the dual has the same characteristic. □

Remark 5.6 (Relation to Brown's theory). Brown's theorem for random walks on left-regular bands [Br00] (and, for hyperplane arrangements, [BHR99, BD98]) indexes the eigenvalues by the support lattice, here the Boolean lattice of subsets X of the cards with $\lambda_X = |X|/N$, and gives the multiplicities $m_X = \sum_{Y \supseteq X} \mu(X, Y)\, (N - |Y|)! = d_{N-|X|}$ through the Möbius function $\mu(X, Y) = (-1)^{|Y \setminus X|}$. Theorem 5.5 is the $\mathfrak{S}_N$-equivariant version: the factor $1/H(t) = E(-t)$ is

the Möbius function with the symmetric group acting, which is why it appears through the sign characters e_i. We used the prefix filtration instead of Brown's idempotent construction because it is visibly $\mathfrak{S}_N$-stable; Saliola [Sal12] gives explicit eigenvectors, which would be the starting point for an idempotent-based equivariant version. The function δ_m is the characteristic of the derangement representation of $\mathfrak{S}_m$ [DW93, RW04], and the module structure in Theorem 5.5 is, we believe, also recorded in [UR02, RSW14]; we have not checked the precise statements there.

5.5. **Step 3: counting invariants.**

Lemma 5.7. *Let V be a representation of $\mathfrak{S}_N$ and $F, F' \in \Lambda$.*

(a) *For a composition μ of N, $\dim V^{\mathfrak{S}_\mu} = \langle \operatorname{ch} V, h_\mu \rangle$.*
(b) *$\langle h_j F, F' \rangle = \langle F, h_j^{\perp} F' \rangle$, where $h_j^{\perp}$ is the adjoint of multiplication by h_j, and $h_j^{\perp} h_\mu = \sum_\alpha h_{\mu-\alpha}$, the sum over $\alpha \in \mathbb{Z}^k_{\geq 0}$ with $\alpha_i \leq \mu_i$ and $|\alpha| = j$. In particular $h_j^{\perp}(h_a h_b) = \sum_{r+g=j} h_{a-r} h_{b-g}$.*
(c) *$\langle F, h_\beta \rangle$ is the coefficient of $x^\beta = x_1^{\beta_1} \cdots x_k^{\beta_k}$ in $F(x_1, \dots, x_k)$.*

Proof. (a) is Frobenius reciprocity: $\dim V^{\mathfrak{S}_\mu} = \dim \operatorname{Hom}(\operatorname{Ind}_{\mathfrak{S}_\mu} \mathbf{1}, V)$. (b) follows from the coproduct $\Delta h_j = \sum_r h_r \otimes h_{j-r}$ and $h_r^{\perp} h_a = h_{a-r}$ [Mac95, Ch. I.5]. (c) holds because $\{h_\lambda\}$ and $\{m_\lambda\}$ are dual bases. □

Proof of Theorem 1.1. By Proposition 5.1 and Theorem 5.5, the eigenvalues of P lie in $\{j/N : 0 \leq j \leq N-2\} \cup \{1\}$, and by Lemma 5.7 the algebraic multiplicity of j/N is

$$\dim V_j(NT)^{\mathfrak{S}_n \times \mathfrak{S}_n} = \langle h_j \delta_{N-j},\, h_n h_n \rangle = \sum_{r+g=j} \langle \delta_{N-j},\, h_{n-r} h_{n-g} \rangle.$$

Only two variables are needed. In two variables $e_1 = x + y$, $e_2 = xy$ and $e_i = 0$ for $i \geq 3$, so

$$\delta_m(x,y) = (x+y)^m - (x+y)(x+y)^{m-1} + xy\,(x+y)^{m-2} = xy\,(x+y)^{m-2} \qquad (m \geq 2),$$

while $\delta_1 = 0$ and $\delta_0 = 1$. The coefficient of $x^a y^b$ in $xy(x+y)^{a+b-2}$ is $\binom{a+b-2}{a-1}$, so $\langle \delta_{a+b},\, h_a h_b \rangle = D(a,b)$ for all $a, b \geq 0$. Hence the multiplicity of j/N is $\sum_{r+g=j} D(n-r, n-g)$, which is $m_n(j)$ by Proposition 4.1(e).

For diagonalizability we use the theorem of Brown and Diaconis [BD98] and Brown [Br00] that random walks on left-regular bands, in particular the Tsetlin library R, are diagonalizable. Then $T = R^{\mathsf{T}}$ is diagonalizable, and so is its restriction P to the invariant subspace $(\mathbb{C}^{\mathfrak{S}_N})^{\Gamma}$ (Proposition 5.1). □

Remark 5.8. The factor xy in $\delta_m(x,y) = xy(x+y)^{m-2}$ is why $D(a,b)$ counts the words that begin with `R` and end with `G` (Remark 1.2), and $\delta_1 = 0$ is why $(N-1)/N$ never occurs: a single card cannot be deranged.

5.6. **Decks of arbitrary content.**

Corollary 5.9. *Let $\mu = (\mu_1, \dots, \mu_k)$ be a composition of N, and consider the top-to-random chain on words with μ_i letters of colour i. It is diagonalizable, its eigenvalues lie in $\{j/N : 0 \leq j \leq N-2\} \cup \{1\}$, and j/N has multiplicity*

$$m_\mu(j) = \langle h_j \delta_{N-j},\, h_\mu \rangle = \sum_{\substack{\alpha \leq \mu \\ |\alpha| = j}} \left[x^{\mu-\alpha}\right] \delta_{N-j}(x_1, \dots, x_k), \qquad \delta_m(x_1, \dots, x_k) = \sum_{i=0}^{k} (-1)^i e_i\, p_1^{m-i}.$$

For two colours, $\mu = (a,b)$ with $a, b \geq 1$, and $0 \leq j \leq N-2$, the multiplicity $m_{(a,b)}(j)$ is the number of words of length $N-2-j$ with at most $a-1$ letters `R` and at most $b-1$ letters `G`.

Proof. Replace Γ by the Young subgroup $\mathfrak{S}_\mu$ in Proposition 5.1 and apply Theorem 5.5 and Lemma 5.7; in k variables $e_i = 0$ for $i > k$. For two colours the computation in the proof of Theorem 1.1 gives $m_{(a,b)}(j) = \sum_{r+g=j} D(a-r, b-g)$. For $j \le N-2$ only the terms with $a-r \ge 1$ and $b-g \ge 1$ survive, and $D(a-r, b-g)$ is the number of words of length $L = N-2-j$ with $a-r-1$ letters `R` and $b-g-1$ letters `G`; summing over r counts the words with at most $a-1$ letters `R` and at most $b-1$ letters `G`. □

Table 5.1 compares Corollary 5.9 with spectra computed directly from the matrices.

TABLE 5.1. Multiplicities of $N/N, (N-1)/N, \dots, 0$ for decks of other contents; the prediction of Corollary 5.9 and the computed spectrum agree in every case.

content μ	states	multiplicities from the top
$(3,2)$	10	1, 0, 1, 2, 3, 3
$(4,2)$	15	1, 0, 1, 2, 3, 4, 4
$(3,3)$	20	1, 0, 1, 2, 4, 6, 6
$(2,2,1)$	30	1, 0, 3, 6, 10, 10
$(3,2,1)$	60	1, 0, 3, 6, 12, 19, 19
$(2,1,1,1)$	60	1, 0, 6, 11, 21, 21

6. How far the formula simplifies

6.1. Two regimes.

Proposition 6.1. *Let $0 \le j \le N-2$ and $L = N-2-j$.*

(i) *If $j \ge n-1$ (equivalently $L \le n-1$), then $m_n(j) = 2^L = 2^{2n-2-j}$.*
(ii) *If $j \le n-2$, then $m_n(j) = 2^L - 2B(L, n-2-j)$: all words of length L minus two tails of $n-1-j$ binomial coefficients each.*
(iii) *For all j, $m_n(j) \ge 2^L\bigl(1 - 2e^{-j^2/(4n)}\bigr)$.*

Proof. (i) A word of length $L \le n-1$ cannot contain n copies of a letter. (ii) By the symmetry of binomial coefficients, $2B(L, n-1) - 2^L = 2^L - 2\sum_{i \ge n}\binom{L}{i} = 2^L - 2\sum_{i \le L-n}\binom{L}{i}$, and $L - n = n-2-j$. (iii) If $L = 0$ then $m_n(j) = 1$ and there is nothing to prove. Otherwise, by Proposition 4.1(b) and Hoeffding's inequality [Ho63], $1 - 2^{-L} m_n(j) = \Pr(|S_L| \ge j+1) \le 2e^{-(j+1)^2/(2L)} \le 2e^{-j^2/(4n)}$, since $L < 2n$. □

So the eigenvalues $(n-1)/N, \dots, (N-2)/N$ have multiplicities $2^{n-1}, \dots, 2, 1$, and the cap of $n-1$ copies per letter only matters noticeably when j is of order $\sqrt{n}$ or smaller.

6.2. The lowest few multiplicities.

Writing out the sums in (1.2) and using Pascal's rule and symmetry gives, for $n \ge 2$,

$$m_n(0) = m_n(1) = \binom{2n-2}{n-1},$$
$$m_n(2) = \binom{2n-2}{n-1} - \binom{2n-4}{n-2} = \frac{3n-5}{n-1}\binom{2n-4}{n-2},$$

and, for $n \geq 3$,

$$m_n(3) = 2\binom{2n-4}{n-3},$$
$$m_n(4) = \frac{5n^2-25n+32}{(n-1)(n-2)}\binom{2n-6}{n-3}.$$

For instance $m_n(1) = \binom{2n-3}{n-2} + \binom{2n-3}{n-1} = \binom{2n-2}{n-1}$, and $m_n(3) = \sum_{i=n-4}^{n-1}\binom{2n-5}{i} = \binom{2n-4}{n-3} + \binom{2n-4}{n-1}$. The equality $m_n(0) = m_n(1)$ is exact, whereas for distinct cards the corresponding multiplicities d_N and Nd_{N-1} differ by ± 1.

For each fixed j, the sum (1.2) consists of $j+1$ binomial coefficients from the middle of row L, all rational multiples of one another, so $m_n(j)$ is a binomial coefficient times a rational function of n, as in the examples above. The rational factor becomes more complicated as j grows (the numerator $5n^2-25n+32$ for $j=4$ is already irreducible over $\mathbb{Q}$), and Theorem 6.5 shows that no single formula of this kind covers all j.

6.3. **Pascal's rule.**

Proposition 6.2. *For $n \geq 1$ and $0 \leq L \leq 2n-2$ put $T(n,L) = m_n(2n-2-L)$, the multiplicity at depth L below the top eigenvalue $(N-2)/N$, and put $T(n,L) = 0$ for $L > 2n-2$. Then $T(n,0) = 1$, and for $n \geq 2$ and $1 \leq L \leq 2n-2$,*

$$T(n,L) = T(n,L-1) + T(n-1,L-1).$$

Equivalently, $m_n(j) = m_n(j+1) + m_{n-1}(j-1)$ for $1 \leq j \leq 2n-3$, and $m_n(0) = m_n(1)$.

Proof. $T(n,L)$ counts the words of length L with at most $n-1$ copies of each letter. Deleting the last letter of such a word gives such a word of length $L-1$. Conversely, a valid word u of length $L-1 \leq 2n-3$ cannot contain $n-1$ copies of both letters, so it can be extended by at least one letter; it can be extended by both letters exactly when it contains at most $n-2$ copies of each, and these words are counted by $T(n-1,L-1)$, which is 0 when $L-1 = 2n-3$. Hence $T(n,L) = T(n,L-1) + T(n-1,L-1)$. □

In Table 6.1 each entry is its left neighbour plus the entry diagonally above and to the left. Each row ends with a repeated entry because the entry above and to the left of the last one is 0; this is the equality $m_n(0) = m_n(1)$.

TABLE 6.1. $T(n,L) = m_n(2n-2-L)$. Column L of row n is the eigenvalue $(2n-2-L)/(2n)$; the eigenvalues 1 (multiplicity 1) and $(2n-1)/(2n)$ (multiplicity 0) are not shown.

	$L=0$	1	2	3	4	5	6	7	8	9	10
$n=1$	1										
$n=2$	1	2	2								
$n=3$	1	2	4	6	6						
$n=4$	1	2	4	8	14	20	20				
$n=5$	1	2	4	8	16	30	50	70	70		
$n=6$	1	2	4	8	16	32	62	112	182	252	252

6.4. **A generating function.**

Proposition 6.3. *Let $C(t) = \big(1-\sqrt{1-4t}\big)/(2t) = \sum_{k\ge 0} C_k t^k$ be the Catalan generating function. Then*

$$\sum_{n\ge 1} t^{n-1} \sum_{j=0}^{2n-2} m_n(j)\, s^j = \frac{1 + s\,tC(t)}{\big(1 - s\,tC(t)\big)\big(1-s^2t\big)\sqrt{1-4t}}.$$

Adding the contribution of the eigenvalue 1*,*

$$\sum_{n\ge 1} s^{2n} t^{n-1} = \frac{s^2}{1-s^2t},$$

gives the generating function $\sum_{n\ge 1} t^{n-1} \sum_\lambda s^{N\lambda}$ of the complete spectra, the inner sum running over the eigenvalues with multiplicity. At $s = 0$ the right-hand side is

$$\frac{1}{\sqrt{1-4t}} = \sum_{n\ge 1} \binom{2n-2}{n-1} t^{n-1},$$

the multiplicities of the eigenvalue 0*; and with the eigenvalue* 1 *included, $s = 1$ gives $\sum_{n\ge 1}\binom{2n}{n} t^{n-1}$, the numbers of states, as it must.*

Proof. A word with a letters `R` and b letters `G` is counted in $m_n(j)$ exactly when $n-1 \ge \max(a,b)$ and $j = 2(n-1) - a - b$. Writing $q = n-1$, the left-hand side is therefore

$$\sum_{a,b\ge 0} \binom{a+b}{a} \sum_{q\ge \max(a,b)} s^{2q-a-b} t^q = \frac{1}{1-s^2t} \sum_{a,b\ge 0} \binom{a+b}{a} s^{|a-b|} t^{\max(a,b)}.$$

The terms with $a = b = q$ give $\sum_q \binom{2q}{q} t^q = 1/\sqrt{1-4t}$. The terms with $|a-b| = d \ge 1$ and $\min(a,b) = q$ come in two orientations and give, by the identity $\sum_{q\ge 0}\binom{2q+d}{q} t^q = C(t)^d/\sqrt{1-4t}$,

$$2\sum_{d\ge 1} (st)^d \frac{C(t)^d}{\sqrt{1-4t}} = \frac{2\,stC(t)}{\big(1-stC(t)\big)\sqrt{1-4t}}.$$

Adding the two contributions gives the formula. The value at $s = 1$ follows from $\big(1-tC(t)\big)C(t) = 1$ and $\sqrt{1-4t} = 1 - 2tC(t)$. □

6.5. **No closed form in the lower half.**

Definition 6.4. Let $d \in \{1,2\}$. A function h on a set $\mathcal{D} \subseteq \mathbb{Z}^2$ is a *hypergeometric term* (with step d in the second variable) if there are rational functions $R_1, R_2 \in \mathbb{Q}(x,y)$ such that $h(n+1,j) = R_1(n,j)\,h(n,j)$ and $h(n,j+d) = R_2(n,j)\,h(n,j)$ whenever both points lie in $\mathcal{D}$.

Products and quotients of factorials of integer-linear forms in n and j, multiplied by rational functions and powers, are hypergeometric terms; this is the usual meaning of a closed form for a single term [PWZ96]. Step $d = 2$ allows separate formulas for even and odd j.

Theorem 6.5. *Let $\mathcal{D} = \{(n,j) \in \mathbb{Z}^2 : n \ge 2,\ 0 \le j \le n-2\}$, the lower half of the spectrum, and for $\varepsilon \in \{0,1\}$ let $\mathcal{D}_\varepsilon = \{(n,j) \in \mathcal{D} : j \equiv \varepsilon \pmod 2\}$. There is no hypergeometric term with step* 2 *on $\mathcal{D}_\varepsilon$ that agrees with $m_n(j)$ there. In particular, no hypergeometric term in (n,j) agrees with $m_n(j)$ on $\mathcal{D}$, even if separate formulas are allowed for even and odd j.*

Proof. Change variables to $p = n-1$ and $L = 2n-2-j$, so that $j = 2p - L$ and $j \equiv L \pmod 2$. The set $\mathcal{D}_\varepsilon$ becomes $\{(p,L) : p \ge 1,\ p+1 \le L \le 2p,\ L \equiv \varepsilon\}$, and by Proposition 4.1(c) the multiplicity becomes

$$M(p,L) = 2B(L,p) - 2^L, \qquad \text{so} \qquad M(p+1,L) - M(p,L) = 2\binom{L}{p+1}.$$

Suppose $h = M$ on $\mathcal{D}_\varepsilon$ for a hypergeometric term h with step 2. The shift $(p, L) \mapsto (p+1, L)$ is $(n, j) \mapsto (n+1, j+2)$, so on $W = \{(p, L) : p+2 \le L \le 2p,\ L \equiv \varepsilon\}$, where both points lie in the domain, $h(p+1, L) = \rho(p, L)\, h(p, L)$ with $\rho = R_1(n, j)R_2(n+1, j)$ rational. On W we have $h(p, L)\big(\rho(p, L) - 1\big) = 2\binom{L}{p+1} > 0$, so

$$h(p, L) = 2\, y(p, L)\binom{L}{p+1}, \qquad y = \frac{1}{\rho - 1} \in \mathbb{Q}(p, L).$$

Let $W' = \{(p, L) : p+3 \le L \le 2p,\ L \equiv \varepsilon\}$, on which (p, L) and $(p+1, L)$ both lie in W. Substituting into $M(p+1, L) - M(p, L) = 2\binom{L}{p+1}$ and dividing by $2\binom{L}{p+1} \neq 0$ gives, on W',

$$(L - p - 1)\, y(p+1, L) - (p+2)\, y(p, L) = p + 2. \tag{6.1}$$

Clearing denominators turns (6.1) into $\Phi(p, L) = 0$ for a polynomial Φ, valid at the points of W' off the zero set of some nonzero polynomial Ψ. Then $\Phi\Psi$ vanishes on W'. For each large p the set W' contains about $p/2$ values of L, more than the degree of $\Phi\Psi$ in L, so every coefficient of $\Phi\Psi$ as a polynomial in L vanishes at every large p, and is therefore zero. Hence $\Phi\Psi = 0$, so $\Phi = 0$ and (6.1) is an identity in $\mathbb{Q}(p, L)$. Now regard L as transcendental over $\mathbb{Q}$ and y as an element of $K(p)$, $K = \mathbb{Q}(L)$, and let $\overline{K}$ be an algebraic closure of K.

y has no poles in $\overline{K}$. Suppose y has a pole α. The poles in $\alpha + \mathbb{Z}$ form a finite nonempty set; let α_+ be its largest element and α_- its smallest. At $p = \alpha_+$ the function $y(p+1)$ is regular, as is the right-hand side of (6.1), so $(p+2)y(p)$ must be regular at α_+, which forces $\alpha_+ = -2$. At $p = \alpha_- - 1$ the function $y(p)$ is regular, so $(L - p - 1)y(p+1)$ must be regular there, which forces $L - \alpha_- = 0$. Then $\alpha_+ - \alpha_- = -2 - L \in \mathbb{Z}$, impossible since L is transcendental.

y is not a polynomial either. If $y \neq 0$ is a polynomial of degree e with leading coefficient c, the left-hand side of (6.1) has leading term $-c\,p^{e+1} - c\,p^{e+1} = -2c\,p^{e+1}$, so $e = 0$ and $y = c$ is constant; then $c(L - 2p - 3) = p + 2$ forces $c = -\frac{1}{2}$ and $L = -1$, a contradiction. And $y = 0$ gives $0 = p + 2$.

This contradiction proves the first statement. For the second, the restriction of a hypergeometric term with step 1 on $\mathcal{D}$ to $\mathcal{D}_\varepsilon$ is a hypergeometric term with step 2. □

Remark 6.6. The argument is Gosper's algorithm [Go78], [PWZ96, Ch. 5] applied to the partial row sums $B(L, p) = \sum_{i \le p} \binom{L}{i}$ of Pascal's triangle, the standard example of a sum with no closed form (in contrast with the alternating sums $\sum_{i \le p}(-1)^i\binom{L}{i} = (-1)^p\binom{L-1}{p}$). So the description of the multiplicities is genuinely piecewise: 2^{2n-2-j} in the upper half $j \ge n - 1$, and in the lower half a partial row sum, for which the recurrence of Proposition 6.2 and the generating function of Proposition 6.3 are the simplest exact descriptions.

7. The spectrum on each Specht module

7.1. **The chain in the group algebra.** Let $\mathfrak{S}_N$ act on words of any content by permuting positions, and on functions by $(\rho(\tau)f)(w) = f(w \circ \tau)$. By (1.1),

$$P = \frac{1}{N}\sum_{p=1}^{N} \rho(\pi_p),$$

the image of an element of the group algebra $\mathbb{C}\mathfrak{S}_N$. Hence P preserves every $\mathfrak{S}_N$-stable subspace, and the multiplicity of an eigenvalue of P on an irreducible submodule depends only on its isomorphism type. For $a \ge b$ write $M^{(a,b)}$ for the functions on words with a letters `R` and b letters `G`; the same element of $\mathbb{C}\mathfrak{S}_N$ acts on $M^{(a,b)}$ as the chain for that deck. By Young's rule [Sag01],

$$M^{(a,b)} \cong \bigoplus_{i=0}^{b} S^{(a+b-i,i)}, \qquad \dim S^{(N-i,i)} = \binom{N}{i} - \binom{N}{i-1},$$

each Specht module occurring once. In particular $\mathbb{C}^{\Omega_n} = M^{(n,n)} \cong S^{(2n)} \oplus S^{(2n-1,1)} \oplus \cdots \oplus S^{(n,n)}$, the top piece $S^{(n,n)}$ has dimension C_n, and $M^{(2n-i,i)} \cong M^{(2n-i+1,i-1)} \oplus S^{(2n-i,i)}$.

7.2. **Multiplicities on the pieces.**

Proposition 7.1. *Let $1 \le i \le n$. On $S^{(2n-i,i)}$ the eigenvalue j/N has multiplicity*

$$\binom{L}{i-1} - \binom{L}{2n-i} \quad (0 \le j \le N-2,\ L = N-2-j),$$

and multiplicity 0 for $j \in \{N-1, N\}$. The trivial module $S^{(2n)}$ carries only the eigenvalue 1.

Proof. By the last isomorphism above, the multiplicity on $S^{(2n-i,i)}$ is $m_{(2n-i,i)}(j) - m_{(2n-i+1,i-1)}(j)$, the difference of the multiplicities for two decks. For $j = N$ both are 1 and for $j = N-1$ both are 0. Let $j \le N-2$. For $i \ge 2$ both decks have two colours, and by Corollary 5.9 the two numbers count the words of length L with at most $2n-i-1$ letters `R` and at most $i-1$ letters `G`, and with at most $2n-i$ letters `R` and at most $i-2$ letters `G`. Counting by the number g of letters `G`, the first count runs over $\max(0, L-2n+i+1) \le g \le i-1$ and the second over $\max(0, L-2n+i) \le g \le i-2$, so the difference is $\binom{L}{i-1} - \binom{L}{L-2n+i} = \binom{L}{i-1} - \binom{L}{2n-i}$. For $i = 1$ the second deck is the single word $\mathtt{R}^{2n}$, with only the eigenvalue 1, and the first count is $1 = \binom{L}{0} - \binom{L}{2n-1}$. Finally $M^{(2n,0)} = S^{(2n)}$ is the trivial module. □

Corollary 7.2. *Let $0 \le j \le N-2$ and $L = N-2-j$.*

(a) $\sum_{i=1}^{n} \left[\binom{L}{i-1} - \binom{L}{2n-i}\right] = B(L, n-1) - \left(2^L - B(L, n-1)\right) = m_n(j)$, *in agreement with Theorem 1.1.*

(b) *The standard representation $S^{(N-1,1)}$ carries each of the eigenvalues $0, 1/N, \ldots, (N-2)/N$ exactly once.*

(c) *On $S^{(n,n)}$ the eigenvalue j/N has multiplicity*

$$\binom{L}{n-1} - \binom{L}{n} = f^{(n-1,\,n-1-j)} \quad (0 \le j \le n-1),$$

and multiplicity 0 for $j \ge n$. Here f^λ is the number of standard Young tableaux of shape λ.

Proof. (a) uses $L \le 2n-2$, so that $\sum_{s=n}^{2n-1} \binom{L}{s} = 2^L - B(L, n-1)$. (b) is the case $i = 1$. (c) is the case $i = n$, together with the hook length formula $f^{(a,b)} = \binom{a+b}{b} - \binom{a+b}{b-1}$ for $a = n-1$, $b = n-1-j$, $a+b = L$, and the symmetry of binomial coefficients. For $j \ge n$ we have $L \le n-2$, so both binomial coefficients vanish. □

TABLE 7.1. $n = 5$: multiplicity of $j/10$ on each Specht module $S^{(10-i,i)}$. The last row is the total, $m_5(j)$, and the last column the dimension.

	$j=0$	1	2	3	4	5	6	7	8	9	10	dim
$S^{(10,0)}$	0	0	0	0	0	0	0	0	0	0	1	1
$S^{(9,1)}$	1	1	1	1	1	1	1	1	1	0	0	9
$S^{(8,2)}$	7	7	6	5	4	3	2	1	0	0	0	35
$S^{(7,3)}$	20	20	15	10	6	3	1	0	0	0	0	75
$S^{(6,4)}$	28	28	19	10	4	1	0	0	0	0	0	90
$S^{(5,5)}$	14	14	9	4	1	0	0	0	0	0	0	42
total	70	70	50	30	16	8	4	2	1	0	1	252

7.3. **An explicit description of the pieces.** Identify a word $w \in \Omega_n$ with the set $G(w) \subseteq [N]$ of positions of its letters $\mathtt{G}$, an n-subset. For $0 \le i \le n$ let U_i send a function ϕ on the i-subsets of $[N]$ to $(U_i\phi)(w) = \sum_{T \subseteq G(w),\,|T|=i} \phi(T)$. Each U_i is $\mathfrak{S}_N$-equivariant and injective (the inclusion matrices have full rank [Got66]), so $\operatorname{Im} U_i \cong M^{(N-i,i)} \cong \bigoplus_{i' \le i} S^{(N-i',i')}$, and the $S^{(N-i,i)}$-component of $\mathbb{C}^{\Omega_n}$ is the orthogonal complement of $\operatorname{Im} U_{i-1}$ in $\operatorname{Im} U_i$ (the standard inner product is invariant, and distinct isotypic components are orthogonal). For $n \le 6$ we computed these components, checked that P preserves each of them, and computed the spectrum of P on each; all agree with Proposition 7.1.

7.4. **The colour swap.**

Lemma 7.3. *Under the identification $w \mapsto G(w)$, the colour swap σ sends an n-subset of positions to its complement. Thus S is the adjacency operator of the Kneser graph $KG(2n, n)$, whose vertices are the n-subsets of $[2n]$, two being adjacent when they are disjoint. It commutes with $\mathfrak{S}_N$ and acts on the component $S^{(2n-i,i)}$ as the scalar $(-1)^i$.*

Proof. Each n-subset of $[2n]$ is disjoint from exactly one other, its complement. The map commutes with permutations of positions, so by Schur's lemma it acts on each (multiplicity-free) component by a scalar. The Kneser graph $KG(N, n)$ has eigenvalue $(-1)^i\binom{N-n-i}{n-i}$ on the $S^{(N-i,i)}$-component [Lo79, GR01], and for $N = 2n$ this is $(-1)^i$. □

Theorem 7.4. *The colour-symmetric and colour-antisymmetric parts of Section 2 are*

$$V_+ = \bigoplus_{i \text{ even}} S^{(2n-i,i)}, \qquad V_- = \bigoplus_{i \text{ odd}} S^{(2n-i,i)}.$$

For $0 \le j \le N-3$ the eigenvalue j/N has the same multiplicity, $m_n(j)/2$, in V_+ and in V_-. The eigenvalue $(N-2)/N$ occurs only in V_- (once), and the eigenvalue 1 only in V_+ (once). In particular $m_n(j)$ is even for $0 \le j \le 2n-3$.

Proof. The description of $V_\pm$ is Lemma 7.3. Let $j \le N-2$. The trivial module carries only the eigenvalue 1, so by Proposition 7.1 the multiplicity of j/N in V_+ minus that in V_- is

$$\Delta = \sum_{i=1}^{n} (-1)^i \Big[\binom{L}{i-1} - \binom{L}{2n-i} \Big].$$

If $L = 0$, that is $j = N-2$, only $i = 1$ contributes (as $2n - i \ge n \ge 1$), so $(N-2)/N$ lies in V_- with multiplicity 1. If $L \ge 1$, use $\sum_{r=0}^{q} (-1)^r\binom{L}{r} = (-1)^q\binom{L-1}{q}$ and $\sum_{r=0}^{L} (-1)^r\binom{L}{r} = 0$:

$$\sum_{i=1}^{n} (-1)^i \binom{L}{i-1} = -(-1)^{n-1}\binom{L-1}{n-1},$$

$$\sum_{i=1}^{n} (-1)^i \binom{L}{2n-i} = \sum_{s=n}^{2n-1} (-1)^s \binom{L}{s} = -(-1)^{n-1}\binom{L-1}{n-1},$$

where the second computation uses $L \le 2n-2$. Hence $\Delta = 0$. The eigenvalue 1 lies in the trivial module, which is in V_+. □

For example, for $n = 5$, Table 7.1 gives the following multiplicities of $j/10$ for $j = 0, 1, \dots, 8$:

$$V_+ :\ 35,\ 35,\ 25,\ 15,\ 8,\ 4,\ 2,\ 1,\ 0, \qquad V_- :\ 35,\ 35,\ 25,\ 15,\ 8,\ 4,\ 2,\ 1,\ 1.$$

For $n = 2$ and $n = 3$ the theorem reproduces the splittings found in Section 3.

8. A proposed formula

It has been suggested that the multiplicity of the eigenvalue $j/(2n)$ is

$$s_n(j) = \binom{2n-j-1}{n-1} - \binom{2n-j-1}{n}, \tag{8.1}$$

first for all j, and in a revised form only for $1 \le j \le 2n-2$. (In the original proposal the eigenvalue numerator was called k.)

Proposition 8.1. *Let $s_n(j)$ be as in (8.1).*

(a) *$s_n(0) = 0$, $s_n(j) = f^{(n-1,\,n-j)}$ for $1 \le j \le n$, and $s_n(j) = 0$ for $n < j \le 2n-1$. The values are ballot numbers, and $\sum_{j=0}^{2n-1} s_n(j) = C_n$.*

(b) *For $1 \le j \le 2n-1$, $s_n(j)$ is the multiplicity of the eigenvalue $(j-1)/(2n)$, one step below $j/(2n)$, on the Specht module $S^{(n,n)}$.*

(c) *As a formula for all j, (8.1) fails for every $n \ge 1$, since $s_n(0) = 0 < m_n(0)$.*

(d) *For $1 \le j \le 2n-2$ we have $s_n(j) \le m_n(j)$, with equality only for $(n,j) = (2,2)$. So the revised claim also fails for every $n \ge 2$ (for $n = 1$ its range is empty).*

Proof. (b) By Corollary 7.2(c), the multiplicity of $(j-1)/N$ on $S^{(n,n)}$ is $\binom{L'}{n-1} - \binom{L'}{n}$ with $L' = 2n-2-(j-1) = 2n-j-1$, which is $s_n(j)$, for $0 \le j-1 \le N-2$.

(a) The hook length formula gives $f^{(n-1,n-j)} = \binom{2n-j-1}{n-j} - \binom{2n-j-1}{n-j-1}$, which equals $s_n(j)$ by symmetry, for $1 \le j \le n$. For $j > n$ the top index $2n-j-1$ is less than $n-1$, so $s_n(j) = 0$; and $s_n(0) = \binom{2n-1}{n-1} - \binom{2n-1}{n} = 0$. By (b) the sum is $\dim S^{(n,n)} = C_n$.

(c) $m_n(0) = \binom{2n-2}{n-1} \ge 1$. For $n = 1$, for example, P has eigenvalues 1 and 0, while (8.1) assigns multiplicity 0 to the eigenvalue 0 and multiplicity 1 to $\frac{1}{2}$.

(d) Compare with $m_n(j)$, using Proposition 6.1 and $L = 2n-2-j$.

- $1 \le j \le n-2$: here $m_n(j)$ is a sum of consecutive binomial coefficients $\binom{L}{i}$ including $i = n-1$ and $i = n-2$, so $m_n(j) \ge \binom{L}{n-1} + \binom{L}{n-2} = \binom{2n-j-1}{n-1}$, while $s_n(j) = \binom{2n-j-1}{n-1} - \binom{2n-j-1}{n}$ and $\binom{2n-j-1}{n} \ge 1$.
- $j = n-1$: $s_n(j) = \binom{n}{n-1} - 1 = n-1 < 2^{n-1} = m_n(j)$.
- $j = n$: $s_n(j) = 1 \le 2^{n-2} = m_n(j)$, with equality only for $n = 2$.
- $n+1 \le j \le 2n-2$: $s_n(j) = 0 < 2^{2n-2-j} = m_n(j)$. □

The smallest counterexample to the revised claim is $n = 2$, $j = 1$: (8.1) gives $\binom{2}{1} - \binom{2}{2} = 1$, but the eigenvalue $\frac{1}{4}$ has the two independent eigenvectors $(1,-2,1,1,-2,1)$ and $(1,-2,1,-1,2,-1)$ of Table 3.1. In general, at $j = 1$ the proposed value is the Catalan number C_{n-1} while the true multiplicity is $\binom{2n-2}{n-1}$. Tables 8.1 and 8.2 show $n = 5$ and $n = 6$.

9. Computational verification

The script in Section A (Python with NumPy; about a minute) checks the following; every check passes.

A. The spectrum of P for $n = 1, \dots, 6$ (up to 924 states) agrees with Theorem 1.1.

B. The spectra for the contents of Table 5.1 agree with Corollary 5.9, including the two-colour word count.

C. For $n = 3$ and $n = 5$: $\dim\ker(NP - jI) = \dim\ker(NP - jI)^2 = m_n(j)$ for every j, in exact integer arithmetic, and these add up to the number of states, so P is diagonalizable; for $n = 5$ the singular values of $10P - jI$ also separate cleanly.

D. For $n = 5$: $\operatorname{tr}\big((10P)^k\big) = \sum_j m_5(j) j^k$ exactly for $0 \le k \le 18$.

E. The matrix $4P$, the eigenvectors of Table 3.1 and the 3×3 blocks for $n = 2$, and the sample rows for $n = 3$.

TABLE 8.1. $n = 5$: the proposed values (8.1), the true multiplicities, and the multiplicities on $S^{(5,5)}$. At $j = 10$ the top index of (8.1) is -1.

j	$s_5(j)$	$m_5(j)$	on $S^{(5,5)}$
0	0	70	14
1	14	70	14
2	14	50	9
3	9	30	4
4	4	16	1
5	1	8	0
6	0	4	0
7	0	2	0
8	0	1	0
9	0	0	0
10	—	1	0
total	42	252	42

TABLE 8.2. $n = 6$, $1 \le j \le 10$: the proposed values $s_6(j) = \binom{11-j}{5} - \binom{11-j}{6}$ and the true multiplicities of $j/12$. The proposed values add up to $C_6 = 132$.

j	$\binom{11-j}{5}$	$\binom{11-j}{6}$	$s_6(j)$	$m_6(j)$
1	252	210	42	252
2	126	84	42	182
3	56	28	28	112
4	21	7	14	62
5	6	1	5	32
6	1	0	1	16
7	0	0	0	8
8	0	0	0	4
9	0	0	0	2
10	0	0	0	1
total			132	

F. For $n \le 6$: the Specht components of Section 7 are P-invariant, have the right dimensions and the spectra of Proposition 7.1; the colour swap acts on them by $(-1)^i$; and the spectra of $P_\pm$ agree as in Theorem 7.4.

G. For $n \le 30$: Proposition 4.1, Proposition 6.1(i),(ii), the formulas for $m_n(0), \dots, m_n(4)$, Proposition 6.2, Corollary 7.2(a), the evenness in Theorem 7.4, and the agreement of (1.2) with $\langle h_j \delta_{N-j},\, h_n h_n \rangle$ for $n \le 8$; and Proposition 6.3 for $n \le 23$.

H. Proposition 8.1: for $n \le 80$ the only agreement of (8.1) with $m_n(j)$ for $1 \le j \le 2n-2$ is $(n, j) = (2, 2)$, and $s_n(j) \le m_n(j)$ throughout; for $n \le 30$, $\sum_j s_n(j) = C_n$ and $s_n(j)$ equals the $S^{(n,n)}$-multiplicity of $(j-1)/N$.

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Note on the references. The bibliographic details above were compiled from memory and should be checked (for instance against MathSciNet) before they are cited. No step of the proofs depends on them except the diagonalizability theorem of [BD98, Br00], which is used only for the word "diagonalizable" in Theorem 1.1 and Corollary 5.9; for $n = 3$ and $n = 5$ diagonalizability was also verified directly.

Appendix A. Verification code

The following script carries out the checks listed in Section 9. It needs only Python 3 and NumPy, and runs in about a minute.

```
#!/usr/bin/env python3
Checks for the note on the top-to-random chain on two-coloured words.

A  spectra for n = 1..6 against the multiplicity formula
B  spectra of other multiset decks against the general-content formula
C  exact eigenspace dimensions for n = 3 and 5 (integer row reduction)
D  traces of powers of 10P for n = 5, k = 0..18, in exact integer arithmetic
E  n = 2 eigenvectors and n = 3 sample rows quoted in the text
F  Specht pieces S^(2n-i,i) and the colour swap, n = 1..6
G  equivalent forms, Pascal's rule, small-j formulas (n <= 30),
   generating function (n <= 23)
H  the suggested formula C(2n-k-1, n-1) - C(2n-k-1, n)
Needs only numpy.  Takes about a minute, mostly for C and F.
```

```
14
15 import itertools
16 from collections import Counter, defaultdict
17 from math import comb, gcd
18 import numpy as np
19
20 R, G = 0, 1
21
22
23 # ------------------------------------------------------------- the chain
24 def chain(content):
25     States (tuples of colour indices) and the integer matrix N*P.
26     N = sum(content)
27     base = [c for c, mult in enumerate(content) for _ in range(mult)]
28     W = sorted(set(itertools.permutations(base)))
29     ix = {w: a for a, w in enumerate(W)}
30     M = np.zeros((len(W), len(W)), dtype=np.int64)
31     for w in W:
32         first, rest = w[0], w[1:]
33         for s in range(N):                      # the N slots
34             M[ix[w], ix[rest[:s] + (first,) + rest[s:]]] += 1
35     return W, ix, M
36
37
38 def spectrum(M):
39     Eigenvalues of the integer matrix M (= N*P), rounded and counted.
40     ev = np.linalg.eigvals(M.astype(float))
41     near = np.rint(ev.real).astype(int)
42     assert np.abs(ev - near).max() < 1e-6
43     return Counter(near.tolist()), np.abs(ev - near).max(), np.abs(ev.imag).max()
44
45
46 # ------------------------------------------------------------ the formulas
47 def m(n, j):
48     Multiplicity of j/(2n) for the chain on words with n R's and n G's.
49     N = 2 * n
50     if j == N:
51         return 1
52     if j == N - 1:
53         return 0
54     L = N - 2 - j
55     return sum(comb(L, i) for i in range(max(0, L - n + 1), min(L, n - 1) + 1))
56
57
58 def polymul(p, q):
59     out = defaultdict(int)
60     for a, x in p.items():
61         for b, y in q.items():
62             out[tuple(u + v for u, v in zip(a, b))] += x * y
63     return out
64
65
66 def delta(deg, k):
67     delta_deg = sum_i (-1)^i e_i p_1^(deg-i) in k variables.
68     p1 = {tuple(int(t == i) for t in range(k)): 1 for i in range(k)}
69     tot = defaultdict(int)
70     for i in range(min(deg, k) + 1):
71         e = {tuple(int(t in S) for t in range(k)): 1
72              for S in itertools.combinations(range(k), i)}
73         p = {(0,) * k: 1}
74         for _ in range(deg - i):
75             p = polymul(p, p1)
76         for ex, c in polymul(e, p).items():
77             tot[ex] += (-1) ** i * c
78     return tot
```

```
79
80
81  def m_content(mu, j):
82      <h_j delta_{N-j}, h_mu>: multiplicity of j/N for a deck of content mu.
83      k, N = len(mu), sum(mu)
84      d = delta(N - j, k)
85      return sum(d.get(tuple(c - a for c, a in zip(mu, al)), 0)
86                 for al in itertools.product(*[range(c + 1) for c in mu])
87                 if sum(al) == j)
88
89
90  def piece(n, i, k):
91      Multiplicity of k/(2n) on the Specht module S^(2n-i,i).
92      N = 2 * n
93      if i == 0:
94          return int(k == N)
95      if k >= N - 1:
96          return 0
97      L = N - 2 - k
98      return comb(L, i - 1) - comb(L, N - i)
99
100
101 def suggested(n, k):
102     t = 2 * n - k - 1
103     return comb(t, n - 1) - comb(t, n)
104
105
106 # ------------------------------------------------------------------------ A
107 print(A. spectra against the formula)
108 for n in range(1, 7):
109     W, ix, M = chain((n, n))
110     obs, dev, im = spectrum(M)
111     assert all(obs.get(j, 0) == m(n, j) for j in range(2 * n + 1))
112     print(   n=%d (%4d states) ok; multiplicities from the top: % (n, len(W)),
113           [m(n, j) for j in range(2 * n, -1, -1)])
114     if n == 5:
115         print(   n=5: max |10*lambda - integer| = %.1e, max |Im lambda| = %.1e
116               % (dev, im / 10))
117
118 # ------------------------------------------------------------------------ B
119 print(B. other decks against <h_j delta_{N-j}, h_mu>)
120 for mu in [(3, 2), (4, 2), (3, 3), (2, 2, 1), (3, 2, 1), (2, 1, 1, 1)]:
121     W, ix, M = chain(mu)
122     N = sum(mu)
123     obs, _, _ = spectrum(M)
124     pred = [m_content(mu, j) for j in range(N, -1, -1)]
125     assert all(obs.get(j, 0) == m_content(mu, j) for j in range(N + 1))
126     if len(mu) == 2:   # two colours: words of length N-2-j, <= a-1 R's, <= b-1 G's
127         a, b = mu
128         for j in range(N - 1):
129             L = N - 2 - j
130             assert m_content(mu, j) == sum(comb(L, g) for g in range(L + 1)
131                                            if g <= b - 1 and L - g <= a - 1)
132     print(   content, mu, ok:, pred)
133
134
135 # ------------------------------------------------------------------------ C
136 def rank_exact(A):
137     Rank over Q of an integer matrix, by integer row reduction.
138     rows = [[int(x) for x in r] for r in A]
139     rank, ncol = 0, len(rows[0])
140     for c in range(ncol):
141         piv = next((i for i in range(rank, len(rows)) if rows[i][c]), None)
142         if piv is None:
143             continue
```

```
        rows[rank], rows[piv] = rows[piv], rows[rank]
        pr, a = rows[rank], rows[rank][c]
        for i in range(rank + 1, len(rows)):
            b = rows[i][c]
            if b:
                row = [a * x - b * y for x, y in zip(rows[i], pr)]
                g = 0
                for x in row:
                    g = gcd(g, x)
                rows[i] = [x // g for x in row] if g > 1 else row
        rank += 1
    return rank


print(C. eigenspace dimensions: exact (n = 3, 5), and by SVD (n = 5))
for n in (3, 5):
    W, ix, M = chain((n, n))
    D = len(W)
    tot, zero_sv, nonzero_sv = 0, 0.0, np.inf
    for j in range(2 * n + 1):
        A = M - j * np.eye(D, dtype=np.int64)
        k1 = D - rank_exact(A)
        k2 = D - rank_exact(A @ A) if k1 else 0
        assert k1 == k2 == m(n, j)
        tot += k1
        sv = np.sort(np.linalg.svd(A.astype(float), compute_uv=False))
        if k1:
            zero_sv = max(zero_sv, sv[k1 - 1])
        nonzero_sv = min(nonzero_sv, sv[k1])
    assert tot == D
    print(   n=%d: dim ker(NP - jI) = dim ker(NP - jI)^2 = m(j) for all j; total %d
          % (n, tot))
print(   n=5 SVD: largest 'zero' singular value %.1e, smallest nonzero %.3f
      % (zero_sv, nonzero_sv))

# ------------------------------------------------------------------------ D
print(D. tr((10P)^k) = sum_j m(j) j^k, n = 5, k = 0..18)
W, ix, M = chain((5, 5))
D = len(W)
Mk = np.eye(D, dtype=np.int64)
for k in range(19):           # entries of M^k are at most 10^k: int64 is exact
    lhs, rhs = int(np.trace(Mk)), sum(m(5, j) * j ** k for j in range(11))
    assert lhs == rhs
    if k in (1, 2, 5, 10, 18):
        print(   k=%2d: %d % (k, lhs))
    Mk = Mk @ M

# ------------------------------------------------------------------------ E
print(E. vectors and rows quoted in the text)
W, ix, M = chain((2, 2))
order = [(R, R, G, G), (R, G, R, G), (R, G, G, R),
         (G, R, R, G), (G, R, G, R), (G, G, R, R)]
P4 = M[np.ix_([ix[w] for w in order], [ix[w] for w in order])]
assert P4.tolist() == [[2, 1, 1, 0, 0, 0], [0, 1, 0, 2, 1, 0], [0, 0, 1, 0, 1, 2],
                       [2, 1, 0, 1, 0, 0], [0, 1, 2, 0, 1, 0], [0, 0, 0, 1, 1, 2]]
for lam4, v in [(4, (1, 1, 1, 1, 1, 1)), (2, (0, 1, -1, 1, -1, 0)),
                (1, (1, -2, 1, 1, -2, 1)), (1, (1, -2, 1, -1, 2, -1)),
                (0, (0, 1, -1, -1, 1, 0)), (0, (1, -2, 0, 0, 2, -1))]:
    assert (P4 @ np.array(v)).tolist() == [lam4 * x for x in v]
Ppl = np.array([[2, 1, 1], [0, 2, 2], [2, 1, 1]])
Pmi = np.array([[2, 1, 1], [0, 0, -2], [-2, -1, 1]])
assert sorted(np.rint(np.linalg.eigvals(Ppl).real).astype(int)) == [0, 1, 4]
assert sorted(np.rint(np.linalg.eigvals(Pmi).real).astype(int)) == [0, 1, 2]
W, ix, M = chain((3, 3))
def row(s):
```

```
    w = tuple(R if ch == 'R' else G for ch in s)
    return {''.join('R' if c == R else 'G' for c in W[b]): int(M[ix[w], b])
            for b in range(len(W)) if M[ix[w], b]}
assert row('RRRGGG') == {'RRRGGG': 3, 'RRGGGR': 1, 'RRGGRG': 1, 'RRGRGG': 1}
assert row('RGRGRG') == {'GRGRGR': 1, 'GRGRRG': 2, 'GRRGRG': 2, 'RGRGRG': 1}
assert row('RRGGGR') == {'RGGGRR': 2, 'RGGRGR': 1, 'RGRGGR': 1, 'RRGGGR': 2}
print(   n=2 matrix, eigenvectors and 3x3 blocks ok; n=3 sample rows ok)


# ------------------------------------------------------------------------ F
def orth(A, tol=1e-9):
    U, s, _ = np.linalg.svd(A, full_matrices=False)
    return U[:, s > tol * s.max()]


print(F. Specht pieces and colour swap)
for n in range(1, 7):
    N = 2 * n
    W, ix, M = chain((n, n))
    D = len(W)
    P = M / N
    swap = np.zeros((D, D))
    for w in W:
        swap[ix[w], ix[tuple(1 - c for c in w)]] = 1
    Gpos = [frozenset(p for p in range(N) if w[p] == G) for w in W]
    Q = [orth(np.array([[1.0 if set(T) <= S else 0.0
                         for T in itertools.combinations(range(N), i)] for S in Gpos]))
         for i in range(n + 1)]
    for i in range(n + 1):
        Wi = Q[i] if i == 0 else orth(Q[i] - Q[i - 1] @ (Q[i - 1].T @ Q[i]))
        assert Wi.shape[1] == comb(N, i) - (comb(N, i - 1) if i else 0)
        A = Wi.T @ P @ Wi
        assert np.abs(P @ Wi - Wi @ A).max() < 1e-9          # P preserves the piece
        assert np.allclose(swap @ Wi, (-1) ** i * Wi)          # swap acts as (-1)^i
        obs, _, _ = spectrum(N * A)
        assert all(obs.get(k, 0) == piece(n, i, k) for k in range(N + 1))
    spec = {}
    for sg in (1, -1):
        B = orth(np.eye(D) + sg * swap)
        spec[sg], _, _ = spectrum(N * (B.T @ P @ B))
    assert all(spec[1].get(k, 0) == spec[-1].get(k, 0) for k in range(N - 2))
    assert (spec[1].get(N - 2, 0), spec[-1].get(N - 2, 0)) == (0, 1)
    assert (spec[1].get(N, 0), spec[-1].get(N, 0)) == (1, 0)
    print(   n=%d ok % n)

# ------------------------------------------------------------------------ G
def Dab(a, b):
    Coloured derangement number D(a,b).
    if a == b == 0:
        return 1
    return 0 if a == 0 or b == 0 else comb(a + b - 2, a - 1)


print(G. equivalent forms, recurrence, generating function)
for n in range(1, 31):
    for j in range(2 * n - 1):
        L = 2 * n - 2 - j
        v = m(n, j)
        assert v == sum(comb(L, i) for i in range(L + 1) if abs(2 * i - L) <= j)
        assert v == 2 * sum(comb(L, i) for i in range(n)) - 2 ** L
        assert v == (2 ** L if j >= n - 1
                     else 2 ** L - 2 * sum(comb(L, i) for i in range(n - 1 - j)))
        assert v == sum(piece(n, i, j) for i in range(1, n + 1))
    for j in range(2 * n + 1):                  # the D(a,b) form
        assert m(n, j) == sum(Dab(n - r, n - (j - r))
```

```
                            for r in range(n + 1) if 0 <= j - r <= n)
        if n <= 8:
            assert m(n, j) == m_content((n, n), j)
    if n >= 2:
        for j in range(2 * n - 2):
            assert m(n, j) == m(n, j + 1) + (m(n - 1, j - 1) if j >= 1 else 0)
        assert m(n, 0) == m(n, 1) == comb(2 * n - 2, n - 1)
        assert m(n, 2) == comb(2 * n - 2, n - 1) - comb(2 * n - 4, n - 2)
        assert m(n, 2) * (n - 1) == comb(2 * n - 4, n - 2) * (3 * n - 5)
    if n >= 3:
        assert m(n, 3) == 2 * comb(2 * n - 4, n - 3)
        assert (m(n, 4) * (n - 1) * (n - 2)
                == comb(2 * n - 6, n - 3) * (5 * n * n - 25 * n + 32))
    for k in range(2 * n - 2):                  # m(k) is even for k <= 2n-3
        assert m(n, k) % 2 == 0

T = 22                                          # series in t, coefficients polys in s
def smul(A, B):
    out = [Counter() for _ in range(T + 1)]
    for i in range(T + 1):
        for k in range(T + 1 - i):
            for e1, c1 in A[i].items():
                for e2, c2 in B[k].items():
                    out[i + k][e1 + e2] += c1 * c2
    return out
cat = [comb(2 * k, k) // (k + 1) for k in range(T + 1)]
u = [Counter() for _ in range(T + 1)]           # u = s t C(t)
for k in range(T):
    u[k + 1][1] = cat[k]
one = [Counter({0: 1})] + [Counter() for _ in range(T)]
ratio, p = [Counter(c) for c in one], one       # (1+u)/(1-u) = 1 + 2u + 2u^2 + ...
for _ in range(T):
    p = smul(p, u)
    for i in range(T + 1):
        for e, c in p[i].items():
            ratio[i][e] += 2 * c
F = smul(smul(ratio, [Counter({2 * r: 1}) for r in range(T + 1)]),
         [Counter({0: comb(2 * k, k)}) for k in range(T + 1)])
for n in range(1, T + 2):
    assert {e: c for e, c in F[n - 1].items() if c} == \
           {j: m(n, j) for j in range(2 * n - 1) if m(n, j)}
print(   identities ok for n <= 30; generating function ok for n <= %d % (T + 1))

# ------------------------------------------------------------------------ H
print(H. the suggested formula)
rng = [(n, k) for n in range(1, 81) for k in range(1, 2 * n - 1)]
hits = [(n, k) for n, k in rng if suggested(n, k) == m(n, k)]
assert hits == [(2, 2)]
assert all(suggested(n, k) <= m(n, k) for n, k in rng)
for n in range(1, 31):
    assert sum(suggested(n, k) for k in range(2 * n)) == comb(2 * n, n) // (n + 1)
    assert suggested(n, 0) == 0
    assert all(suggested(n, k) == piece(n, n, k - 1) for k in range(1, 2 * n))
for n in (5, 6):
    print(   n=%d, k=1..%d: suggested % (n, 2 * n - 2),
          [suggested(n, k) for k in range(1, 2 * n - 1)],
           actual, [m(n, k) for k in range(1, 2 * n - 1)])
print(   agreements for 1 <= k <= 2n-2, n <= 80:, hits)
print(all checks passed)
```